

Response Letter with Annotations

Citation (APA format, using Zotero or other bibliography software)

Van Atteveldt, Wouter, Mariken ACG Van der Velden, and Mark Boukes. "The validity of sentiment analysis: Comparing manual annotation, crowd-coding, dictionary approaches, and machine learning algorithms." *Communication Methods and Measures* 15.2 (2021): 121-140.

Annotations

1. Purposes

a. Motivation (one-sentence summary)

The goal of the study was to compare different methods of conducting sentiment analysis in a communication context, specifically in a non-English language (Dutch).

b. Specific Rationale (several bullet points)

- To address validation issues in using automatic sentiment analysis methods
- To examine various dictionaries
- To test automatic sentiment analysis in non-English language
- To compare manual and automated sentiment analysis against set-standard
- To develop step-by-step approach for automated text analysis

c. Research Questions and/or Hypotheses

- Identify IV, DV, mediator, moderator
- Identify specific relationships among variables

Note: Sentiment: positive, negative or neutral

2. Method

a. Research Design

- Comparative validation. Trainer coders, crowd coders, dictionary and machine learning.
- Triple-coded gold standard validation. 300 headlines.

b. Sample (representative sample?)

- Data collected in previous study (Boukes et al. (2020))
- News headlines in Dutch language, economic sector (likely to be objective).
- Not representative

c. Measures

- Manual coding: Coders assign sentiment. 22 trained students coded. This sentiment measurement proved to have satisfactory inter-coder reliability (Krippendorff's $\alpha = .80$).
- Crowd coding: Five crowd coders recruited on [figure-eight.com](https://www.figure-eight.com) platform
- Dictionary: words from dictionary. 6 Dutch dictionaries & 9 English dictionaries were used. Google Translate and DeepL were used to translate from Dutch to English.
- Machine learning: lemmatized words. Classical: NB and SVM. Deep: CNN.

d. Analysis

- Analysed against gold-standard validation
- Error analysis was done for dictionary and machine-learning methods

3. Results and Conclusions

a. Answers to RQs and Hypotheses

Across the four methods, manual annotation performed best, with three trained coders reaching a Krippendorff's α above .90 and single-coder manual annotation achieving $\alpha = .82$. Crowd coding performed nearly as well, with majority voting across multiple coders reaching $\alpha \geq .80$, comparable to single trained student coding. Machine learning methods performed substantially worse than human coding but better than dictionaries, with deep learning (CNN) achieving $\alpha = .50$ and classical models such as SVM and Naïve Bayes performing slightly lower ($\alpha \approx .40$). Dictionary-based approaches performed worst overall, with most alphas close to chance agreement and the best-performing dictionary reaching only $\alpha = .34$. Overall, the results indicate that human coding remains the most valid method, crowd coding offers a viable and more affordable alternative, machine learning shows moderate performance, and dictionary approaches demonstrate limited validity in this context.

b. Include Key Figures/Tables

Table 2. Overall performance of the tested sentiment analysis approaches.

section	name	Acc.	alpha	Positive			Neutral			Negative		
				Pr.	Re.	F1	Pr.	Re.	F1	Pr.	Re.	F1
Manual Coding	Single Coder	0.82	0.82	0.88	0.86	0.87	0.76	0.81	0.78	0.84	0.80	0.82
Manual Coding	Vote (3 Coders)	0.88	0.90	0.97	0.91	0.94	0.82	0.88	0.85	0.87	0.84	0.86
Crowd-Coding	Single Coder	0.72	0.75	0.69	0.84	0.76	0.69	0.58	0.63	0.78	0.78	0.78
Crowd-Coding	Vote (3 Coders)	0.77	0.81	0.73	0.89	0.80	0.74	0.65	0.69	0.83	0.81	0.82
Crowd-Coding	Vote (5 Coders)	0.77	0.81	0.73	0.90	0.81	0.73	0.65	0.69	0.84	0.80	0.82
Machine Learning	CNN	0.63	0.50	0.68	0.49	0.56	0.58	0.78	0.66	0.72	0.57	0.63
Machine Learning	NB	0.58	0.39	0.74	0.34	0.47	0.52	0.83	0.64	0.65	0.47	0.55
Machine Learning	SVM	0.57	0.41	0.69	0.37	0.48	0.52	0.79	0.62	0.64	0.48	0.55
Dictionaries	DANEW	0.42	0.10	0.75	0.08	0.15	0.40	0.97	0.57	0.80	0.04	0.08
Dictionaries	DamstraBoukes	0.41	0.05	0.83	0.07	0.13	0.40	0.99	0.57	0.00	0.00	0.00
Dictionaries	Muddiman	0.49	0.31	0.53	0.38	0.44	0.46	0.64	0.53	0.53	0.39	0.45
Dictionaries	NRC	0.47	0.32	0.39	0.53	0.45	0.46	0.44	0.45	0.59	0.46	0.52
Dictionaries	Pattern	0.39	0.07	0.43	0.08	0.14	0.39	0.90	0.54	0.38	0.03	0.06
Dictionaries	Polyglot	0.42	0.26	0.38	0.32	0.34	0.39	0.55	0.45	0.53	0.33	0.41
English Dictionaries	AFINN	0.43	0.27	0.35	0.38	0.37	0.40	0.50	0.45	0.58	0.38	0.46
English Dictionaries	DamstraBoukes	0.42	0.07	0.67	0.08	0.15	0.40	0.98	0.57	1.00	0.02	0.04
English Dictionaries	GenInq	0.41	0.26	0.31	0.37	0.34	0.38	0.38	0.38	0.54	0.47	0.51
English Dictionaries	HuLiu	0.46	0.34	0.40	0.30	0.34	0.42	0.63	0.50	0.65	0.40	0.50
English Dictionaries	LoughranMcDonald	0.50	0.29	0.50	0.14	0.22	0.46	0.79	0.58	0.62	0.43	0.51
English Dictionaries	LSD	0.46	0.33	0.39	0.40	0.39	0.42	0.54	0.48	0.62	0.41	0.50
English Dictionaries	Muddiman	0.48	0.27	0.48	0.38	0.43	0.46	0.71	0.55	0.57	0.30	0.39
English Dictionaries	NRC	0.42	0.23	0.34	0.62	0.44	0.43	0.32	0.37	0.57	0.39	0.46
English Dictionaries	RID	0.42	0.06	0.00	0.00	0.00	0.41	0.97	0.57	0.82	0.09	0.16

Acc.: Accuracy expressed as percentage correct; α : Krippendorff's alpha (ordinal), Pr.: Precision for the specified class (Positive/Neutral/Negative); Re.: Recall for that class; F1: F1-Score for that class. When more than one measurement or prediction (respectively for manual/crowd annotation and machine learning) was available, we averaged the score.

c. Authors' Conclusions

Dictionary coding is probably the most used automatic sentiment analysis method in the social sciences due to its transparency and simplicity. Unfortunately, but not surprisingly given previous results (Boukes et al., 2020; González-Bailón & Paltoglou, 2015; Soroka et al., 2015), performance of dictionaries was not satisfactory. Agreement was generally close to chance agreement, and correlations between dictionaries were also low. Error analysis showed that this

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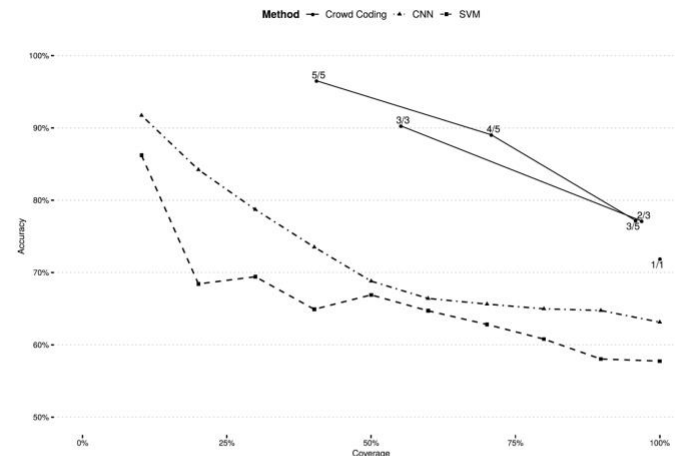


Figure 1. Coverage vs accuracy for crowd coding and machine learning.

Note: For Machine Learning (CNN and SVM), points show the percentage accuracy on the subset of documents on which the algorithm was most confident. For example, the second dash-dotted triangle shows that the CNN model was accurate in about 84% of cases for the 20% of documents on which it was most confident. For Crowd Coding, the lines indicate how many coders were used, and each point represents a majority vote of at least that size. For example, the point '4/5' shows that if at least 4 out of 5 coders agreed on a label, they were correct in almost 90% of cases (accuracy), but only just under 75% of documents were coded with this majority (coverage).

mostly due to the
context of words. This
for both Dutch
dictionaries and for
dictionaries applied to
automatically
translated texts.

In line with
from the
computational
linguistics community,
learning significantly
outperformed
dictionary coding,
“deep learning”
convolutional neural

network scoring around 20 percentage points higher on both agreement and intercoder reliability measures (Liu, 2012; Rosenthal et al., 2017; Rudkowsky et al., 2018). This performance is notwithstanding the fact that a relatively low amount of training material was available, implying that better results might still be achievable with more training data.

4. Discussion

a. Implications

b. Limitations

- Translation tools may not have captured accurate meaning of Dutch words
- Headlines are too short for dictionary analysis
- Sentiment is hard to define, conceptualize and measure

c. Future Directions

d. Recommendations:

There are many different techniques for sentiment analysis and classifications of texts more generally. Unfortunately, the results of this article show that automatic techniques do not always perform sufficiently. This even holds for dictionaries that have been developed and validated independently: Since tasks and domains are almost never interchangeable in social

science, validated performance on the task a tool was developed for does not guarantee sufficient performance on a new task.

On the basis of these findings, we recommend that text classification projects should follow the following steps to guarantee validity:

1. Formalize the concept and operationalization for manual annotation of the construct of interest, often through pilot coding and discussion.
2. Create a sufficiently large gold standard dataset, coded by at least two annotators to assess intercoder reliability.
3. Test off-the-shelf dictionaries against the gold standard. If one performs sufficiently well, use it for analysis.
4. If necessary, customize or build a dictionary, ensuring the gold standard is not used in its construction to avoid bias. Validate it against the gold standard.
5. If dictionaries fail, move to human coding or machine learning, coding a larger dataset (e.g., 1,000+ texts).
6. Train and validate a machine learning model, using cross-validation and testing against the gold standard.
7. Iterate from Step-5 if needed, repeating coding and model refinement until sufficient validity is achieved.

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Response Letter (~600 words)

- Your assessment of the study
- Weaknesses
- Other notes

This study compared manual and automated sentiment analysis methods using Dutch economic news headlines. Overall, the study illustrates the value of human coding when dealing with sentiment, especially in contexts where meaning can be subject to interpretation. The results suggest that trained human coders and even crowd coders perform more effectively than dictionary-based approaches, and machine learning models still do not reach the same level of reliability. At the same time, the authors do not argue that automated methods should be avoided. Instead, they emphasize that automated approaches need to be carefully validated and adapted for research.

One of the strengths of this study is that it does not only compare methods and pick one method over another. The authors use a triple-coded gold standard dataset and systematically evaluate manual coding, crowd coding, dictionary approaches, and machine learning models. They also report reliability statistics such as Krippendorff's alpha to compare across methods. In addition, the authors explain why dictionary approaches fail in this context, pointing to issues such as missing context, negation, and low recall.

I believe there are some limitations in this study. First, the study is based on Dutch-language economic headlines. The authors mention that automated tools may not be as developed for non-English languages, which could partly explain the weak performance of dictionaries and machine learning. At the same time, this is also what makes the study important, because many communication studies rely on English-based tools even when working in other languages. This suggests that non-English languages may require more tailored dictionaries and machine learning approaches for effective sentiment analysis.

Another limitation can be the nature of the data. Headlines are extremely short and often not full sentences. They may lack context and sometimes rely on implicit meaning. While the authors mention that automated methods are often used for short texts like tweets, I would argue that headlines can be even more compressed and ambiguous than tweets. This might make them particularly difficult for dictionary-based and machine learning approaches. It is possible that longer texts would yield different results. Also, the classification into only three categories (positive, negative, neutral) may have made the task manageable for crowd coders with minimal training. I also think that comparing human judgment to dictionary-based word matching naturally creates a gap. Human coders interpret meaning, tone, and context, while dictionaries simply count words. This difference in conceptual approach may partly explain why dictionary performance was close to chance levels. Machine learning performs better because it can capture patterns beyond individual words, but even then, it still depends heavily on the quality and size of the training data. The authors also note that deep learning is less transparent than other methods, and even then it performed at only about 50% agreement with the gold standard.

What I found useful in the article is the seven-step validation procedure the authors propose. Rather than recommending one method over another, they suggest formalizing the concept, creating a gold standard, testing dictionaries, customizing them if needed, and then moving to machine learning only if necessary. I agree with the idea that validation should come

before substantive analysis. However, I am not fully convinced that this workflow fits every study. The choice of method depends on the research question, the size of the dataset, the theoretical definition of sentiment, and the language context. Therefore, this study serves as a reminder that automated methods should be used with appropriate validation to ensure stronger and more reliable results.